An Independent Method for Deriving Closed Forms of Homogeneous Second-Order Linear Recurrence Relations With Constant Coefficients

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July 2026

Abstract

This paper investigates a method for deriving closed forms of homogeneous second-order linear recurrence relations with constant coefficients. It uses standard methods for solving differential equations relating these differential equations to recurrence relations to construct these closed forms. This method was developed independently, as part of this investigation, without referencing existing literature.

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1 Introduction

1.1 Linear recurrence relations

1.1.1 General linear recurrence relations

A linear recurrence relation is an equation that defines each term of a sequence as a linear combination of one or more preceding terms, possibly with coefficients that depend on the index n, and an additional term independent of the sequence. The general form of a linear recurrence relation of order k is:

$$a_n = c_1(n)a_{n-1} + c_2(n)a_{n-2} + \cdots + c_k(n)a_{n-k} + g(n)$$

Where:

- a_n is the n-th term of the sequence,
- $c_1(n), c_2(n), \ldots, c_k(n)$ are coefficient functions that may depend on n,
- $g(n)$ is a function independent of the sequence terms and represents the non-homogeneous component,
- Initial values of $a_1, a_2, \ldots, a_k$ are required to uniquely determine the sequence.

An example of a linear recurrence relation could be:

$$a_n = n^2 \cdot a_{n-1} + 2a_{n-2} + cos(\pi n) + 1$$

Where:

- $a_1 = 0$
- $a_2 = 1$

Defining the sequence:

$$0, 1, 9, 148, 3718, \ldots$$

1.1.2 Homogeneous linear recurrence relations with constant coefficients

A homogeneous linear recurrence relation with constant coefficients is a linear recurrence relation where each term of a sequence is defined as a linear combination, of one or more preceding terms, with constant coefficients and no term independent of the sequence. The general form of a homogeneous linear recurrence relation of order k is:

$$a_n = c_1a_{n-1} + c_2a_{n-2} + \cdots + c_ka_{n-k}$$

Where:

- a_n is the n-th term of the sequence,
- $c_1, c_2, \ldots, c_k$ are constant coefficients,
- Initial values of $a_1, a_2, \ldots, a_k$ are required to uniquely determine the sequence.

An example of a linear recurrence relation could be:

$$a_n = a_{n-1} - 2a_{n-2}$$

Where:

- $a_1 = 1$
- $a_2 = 1$

Defining the sequence:

$$1, 1, -1, -3, -1, 5, \ldots$$

In this paper, I will primarily investigate these homogeneous linear recurrence relations with constant coefficients which have order $k = 2$. These have the general form:

$$a_n = c_1 a_{n-1} + c_2 a_{n-2}$$

Where $c_2 \neq 0$.

1.2 Closed forms of recurrence relations

1.2.1 Definition of closed forms

A closed form of a recurrence relation is an explicit formula that defines the n-th term of the sequence directly in terms of n without requiring any previous terms of the sequence. So, the recurrence relation can be written as:

$$a_n = f(n)$$

Where:

- a_n is the n-th term of the sequence,
- $f(n)$ gives the value of the n-th term directly, taking only n as input.

1.2.2 Computational efficiency

Closed forms are useful when computing any term in a sequence as they allow for a direct calculation without computing previous terms which significantly speeds up computation. For example, in terms of the Big O notation, calculating the n-th term of a sequence defined by a recurrence relation, by computing all previous terms, would generally have a time complexity of $O(n)$. This means that the time taken to compute the n-th term in the sequence will grow proportionally to n, so for large n this computation may be inefficient. However, directly calculating the n-th term of a sequence defined by a recurrence relation using a closed form function would generally have a time complexity of $O(1)$. This means that the time taken to compute the n-th term would remain constant regardless of n, so that this computation will be as efficient for large n as it would for small n. Clearly, this implies how the calculation of these terms using closed forms can greatly improve computational efficiency, especially in regard to large n.

1.2.3 Other applications of closed forms

Closed forms of recurrence relations are also useful because they make it easier to:

- Analyse the behaviour of a sequence,
- Compare sequences and identify patterns,
- Prove properties about sequences.

1.3 The independency of this investigation

I have not read any mathematical literature relating to, or have been educated on, methods for deriving closed forms of recurrence relations. I have recently completed A-Level Mathematics and Further Mathematics, where I was educated in solving up to second-order homogeneous and non-homogeneous differential equations, and the fundamental conception of recurrence relations. I have developed this method completely independently by applying these topics which I have been educated on. This method is likely to already be used and understood, or have similarities to existing methods. However, I am writing this paper purely to practice writing scientific mathematical papers as I aspire to be a experienced mathematician and am enthusiastic to write about the method which I have discovered independently. Although I have not researched methods of deriving closed forms of recurrence relations, I have researched the definitions and types of recurrence relations, such as order, homogeneity, and linearity. I carried out this research after discovering my methods for deriving closed forms, so that I could provide more rigorous and compendious definitions of the certain types of recurrence relations which I will be deriving closed forms of.

2 The Method

2.1 Initial conception

2.1.1 The Fibonacci sequence

Consider the Fibonacci sequence, defined by the recurrence relation:

$$a_n = a_{n-1} + a_{n-2}$$

Where:

- $a_1 = 1$
- $a_2 = 1$

Therefore, defining the Fibonacci sequence to be:

$$1, 1, 2, 3, 5, 8, 13, \ldots$$

Let's say that we want to define a function $f(x)$, where the n-th derivative of $f(x)$ where $x = 0$ is equal to a_n, so that $f^n(0) = a_n, \;\; \forall n \in \mathbb{N}$. My intuition for this idea came from experimenting with Maclaurin series. Therefore, $f(x)$ can be defined to satisfy $f^n(x) = f^{n-1}(x) + f^{n-2}(x)$. In turn, this would mean that $f''(x) = f'(x) + f(x)$, giving us a second-order, homogeneous differential equation which can be rearranged:

$$f''(x) - f'(x) - f(x) = 0$$

Assume a solution of the form $f(x) = Ce^{mx}$ where C and m are constants. Therefore, $f'(x) = mCe^{mx}$ and $f''(x) = m^2Ce^{mx}$ which gives us the equation:

$$m^2Ce^{mx} - mCe^{mx} - Ce^{mx} = 0$$

Dividing both sides by Ce^{mx} gives us the auxiliary equation:

$$m^2 - m - 1 = 0$$

Simply using the quadratic formula gives us the possible values for m:

$$m = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(-1)}}{2(1)}$$

$$\therefore \;\; m = \frac{1 \pm \sqrt{5}}{2}$$

So, we can derive the general solution to the differential equation:

$$f(x) = Ae^{\frac{1+\sqrt{5}}{2}x} + Be^{\frac{1-\sqrt{5}}{2}x}$$

If $a_1 = 1$ and $a_2 = 1$, then $a_0 = 0$ is valid, as $a_2 = a_1 + a_0 = 1 + 0 = 1$ as required. So, as $f^n(0) = a_n$, we can determine that $f(0) = a_0$ which gives us $f(0) = 0$. Substituting this into our general solution, we get:

$$0 = A + B \implies B = -A$$

Differentiating the general solution gives us:

$$f'(x) = \frac{1+\sqrt{5}}{2}Ae^{\frac{1+\sqrt{5}}{2}x} + \frac{1-\sqrt{5}}{2}Be^{\frac{1-\sqrt{5}}{2}x}$$

Using $f^n(0) = a_n$ again, we can determine that $f'(0) = 1$. Therefore:

$$1 = \frac{1+\sqrt{5}}{2}A + \frac{1-\sqrt{5}}{2}B$$

$$\therefore\ B = -A \implies 1 = \frac{1+\sqrt{5}}{2}A + \frac{\sqrt{5}-1}{2}A$$
$$\implies 1 = \sqrt{5}A$$
$$\implies A = \frac{1}{\sqrt{5}} \text{ and } B = -\frac{1}{\sqrt{5}}$$

Which gives us the particular solution:

$$f(x) = \frac{1}{\sqrt{5}}\left(e^{\frac{1+\sqrt{5}}{2}x} - e^{\frac{1-\sqrt{5}}{2}x}\right)$$

Simply finding the n-th derivative gives us:

$$f^n(x) = \frac{1}{\sqrt{5}}\left(\left(\frac{1+\sqrt{5}}{2}\right)^n e^{\frac{1+\sqrt{5}}{2}x} - \left(\frac{1-\sqrt{5}}{2}\right)^n e^{\frac{1-\sqrt{5}}{2}x}\right)$$

Finally, as $f^n(0) = a_n$, this gives us the formula for the n-th term of the Fibonacci sequence:

$$a_n = \frac{1}{\sqrt{5}}\left(\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right)$$

To verify this, I have researched the closed form of the Fibonacci sequence and found my formula to be valid and that it is known as Binet's formula.

2.1.2 An oscillatory example

Consider again the example recurrence relation previously used when defining homogeneous linear recurrence relations with constant coefficients, which was defined as:

$$a_n = a_{n-1} - 2a_{n-2}$$

Where:

- $a_1 = 1$
- $a_2 = 1$

Defining the sequence:

$$1, 1, -1, -3, -1, 5, \ldots$$

Which clearly oscillates, as the sign of the terms changes repeatedly. Let's again define a function $f(x)$, so that $f^n(0) = a_n$, $\forall n \in \mathbb{N}$. Therefore, we can define $f(x)$ to satisfy $f^n(x) = f^{n-1}(x) - 2f^{n-2}(x)$. Rearranging gives us a second-order homogeneous differential equation which $f(x)$ satisfies:

$$f''(x) - f(x) + 2f(x) = 0$$

Assuming an exponential solution, as we did before, gives us the auxiliary equation:

$$m^2 - m + 2 = 0$$

Again, simply using the quadratic formula gives us the possible values of the constant m:

$$m = \frac{1 \pm \sqrt{7}i}{2}$$

Giving us the general solution:

$$f(x) = Ae^{\frac{1+\sqrt{7}i}{2}x} + Be^{\frac{1-\sqrt{7}i}{2}x}$$

Using Euler's identity, this can be rearranged to give us the general solution:

$$f(x) = e^{\frac{1}{2}x}\left((A+B)\cos\left(\frac{\sqrt{7}}{2}x\right) + i\,(A-B)\sin\left(\frac{\sqrt{7}}{2}x\right)\right)$$

Which can be rewritten as:

$$f(x) = e^{\frac{1}{2}x}\left(P\cos\left(\frac{\sqrt{7}}{2}x\right) + Q\sin\left(\frac{\sqrt{7}}{2}x\right)\right)$$

As $a_1 = 1$ and $a_2 = 1$, $a_0 = 0$ is valid as $a_2 = a_1 - 2a_0 = 1 - 2(0) = 1$. So, as $f^n(0) = a_n$, we can determine that$f(0) = a_0$. Substituting this into our general solution, we get:

$$0 = P$$

Because of this, we know that the particular solution is in the form:

$$f(x) = e^{\frac{1}{2}x}\left(Q\sin\left(\frac{\sqrt{7}}{2}x\right)\right)$$

Differentiating this using the product rule gives us:

$$f'(x) = e^{\frac{1}{2}x}\left(\frac{\sqrt{7}}{2}Q\cos\left(\frac{\sqrt{7}}{2}x\right) + \frac{1}{2}Q\sin\left(\frac{\sqrt{7}}{2}x\right)\right)$$

$f'(0) = a_1 = 1$. Therefore:

$$1 = \frac{\sqrt{7}}{2}Q \implies Q = \frac{2}{\sqrt{7}}$$

Which gives us the particular solution:

$$f(x) = e^{\frac{1}{2}x}\left(\frac{2}{\sqrt{7}}\sin\left(\frac{\sqrt{7}}{2}x\right)\right)$$

To find the n-th derivative of $f(x)$, I will use Leibniz's theorem which shows that for $y = uv$:

$$\frac{d^n y}{dx^n} = \sum_{r=0}^{n} \binom{n}{r} \frac{d^r y}{dx^r}\frac{d^{n-r} y}{dx^{n-r}}$$

$$\therefore \frac{d^n y}{dx^n} = \sum_{r=0}^{n} \frac{n!}{r!(n-r)!}\frac{d^r y}{dx^r}\frac{d^{n-r} y}{dx^{n-r}}$$

To use this, we will define the u and v of $f(x)$:

$$u := e^{\frac{1}{2}x} \implies \frac{d^n u}{dx^n} = \left(\frac{1}{2}\right)^n e^{\frac{1}{2}x}$$

$$v := \frac{2}{\sqrt{7}}\sin\left(\frac{\sqrt{7}}{2}x\right)$$

As $\frac{d^n}{dx^n}(\sin kx) = k^n \sin\left(kx + \frac{n\pi}{2}\right)$, this gives us:

$$\frac{d^n v}{dx^n} = \left(\frac{\sqrt{7}}{2}\right)^{n-1}\sin\left(\frac{\sqrt{7}}{2}x + \frac{n\pi}{2}\right)$$

Therefore, applying Leibniz's theorem gives us:

$$f^n(x) = \sum_{r=0}^{n}\frac{n!}{r!(n-r)!}\cdot\left(\frac{1}{2}\right)^r e^{\frac{1}{2}x}\cdot\left(\frac{\sqrt{7}}{2}\right)^{n-r-1}\sin\left(\frac{\sqrt{7}}{2}x + \frac{(n-r)\pi}{2}\right)$$

Therefore, as $f^n(0) = a_n$:

$$a_n = \sum_{r=0}^{n}\frac{n!}{r!(n-r)!}\cdot\left(\frac{1}{2}\right)^r\cdot\left(\frac{\sqrt{7}}{2}\right)^{n-r-1}\sin\left(\frac{(n-r)\pi}{2}\right)$$

Giving us the closed form of this oscillatory recurrence relation. Through some empirical testing, I have checked the first 6 terms and confirmed their validity.

2.1.3 Brief overview

In essence, to solve homogeneous second-order linear recurrence relations with constant coefficients, I define a function $f(x)$ where $f^n(x)$ satisfies the same equation as the n-th term in the recurrence relation. This equation satisfied by $f^n(x)$ is a differential equation. By using standard methods for solving differential equations, I derive a general solution to $f(x)$. By analysing this general solution of $f(x)$, I determine a general expression for any $f^n(x)$. I then set $f^n(k) = a_n$, where a_n is the n-th term in the sequence defined by the recurrence relation and k is a constant. I then derive a particular solution of $f^n(k)$, hence giving me a closed form expression for a_n. When setting $f^n(k) = a_n$, there is no requirement to set $k = 0$, however, this is typically the simplest k to work with, so I generally use it. I will generally derive the value of what a_0 would be even though a_0 is not defined in the initial conditions, and is technically not defined to be within the sequence. However, it holds all the same properties as sequence terms, so can be used in this case, and is found so that I can use a_0 and a_1 rather than a_1 and a_2. I do this because deriving a_0 is fairly simple, whereas differentiating $f(x)$ twice in order to use a_2 would likely require more computation.

2.2 Formulas for specific cases

2.2.1 Derivations

To speed up the computational efficiency of my method for deriving these closed forms of homogeneous second-order linear recurrence relations with constant coefficients, I will derive formulas which can be used directly to find a closed-form of any of these recurrence relations. All of these recurrence relations will have the form:

$$a_n = -c_1 a_{n-1} - c_2 a_{n-2}$$

Where c_1 and c_2 are constants, $c_2 \neq 0$, and the initial values of a_1 and a_2 have been specified. Giving the corresponding auxiliary equation:

$$m^2 + c_1 m + c_2 = 0$$

Where m is the variable whose values are the roots of the equation. Finding these roots helps me to derive the closed form of the recurrence relation. The nature of these roots impacts the structure of this closed form. Because of this, I will derive different specific case formulas dependent on the nature of these roots. Throughout the following derivations, I will assume that I have already defined the recurrence relation with a_n being the n-th term in the sequence determined by this recurrence relation,and set the values of a_1 and a_2 as required. I will also assume I have derived the auxiliary equation with m as the variable by assuming an exponential solution, and defined the function $f(x)$ where $f^n(x)$ satisfies the same equation (which is the recurrence relation) as a_n. Additionally, I will assume that the value of a_0 has been determined:

$$a_2 = -c_1a_1 - c_2a_0 \implies a_0 = -\frac{a_2 + c_1a_1}{c_2}$$

Case 1
I will define case 1 to be the case where:

$$c_1^2 - 4c_2 > 0$$

In this case, there will be 2 real, distinct roots of the auxiliary equation (deduced using the discriminant). Let these 2 roots be denoted by α and β. Therefore, as a solution in the form $f(x) = Ce^{mx}$ would have been assumed, we can derive the general solution of $f(x)$ to be:

$$f(x) = Ae^{\alpha x} + Be^{\beta x}$$

Where A and B are constants. Setting $f^n(0) = a_n$ lets us deduce that:

$$a_0 = A + B \implies B = a_0 - A$$

Differentiating $f(x)$ gives us:

$$f'(x) = \alpha Ae^{\alpha x} + \beta Be^{\beta x}$$

Therefore, letting us deduce that:

$$a_1 = \alpha A + \beta B \implies A = \frac{a_1 - \beta a_0}{\alpha - \beta} \quad (\text{As } B = a_0 - A)$$

In turn, allowing us to determine that:

$$B = \frac{a_0(\alpha - \beta) - (a_1 - \beta a_0)}{\alpha - \beta}$$

$$\therefore\ B = \frac{\alpha a_0 - a_1}{\alpha - \beta}$$

This gives us our particular solution of $f(x)$:

$$f(x) = \frac{1}{\alpha - \beta}\Big((a_1 - \beta a_0)\, e^{\alpha x} + ((\alpha a_0 - a_1)\, e^{\beta x}\Big)$$

Simply analysing $f(x)$, and using the fact that $f^n(0) = a_n$, allows us to derive our closed form equation for case 1 recurrence relations:

$$a_n = \frac{1}{\alpha - \beta}\left((a_1 - \beta a_0)\,\alpha^n + ((\alpha a_0 - a_1)\,\beta^n\right) \tag{1}$$

Case 2

I will define case 2 to be the case where:

$$c_1^2 - 4c_2 = 0$$

In this case, there will be 1 real, distinct root of the auxiliary equation. Let this root be denoted by α. Therefore, as a solution in the form $f(x) = Ce^{mx}$ would have been assumed, we can derive the general solution of $f(x)$:

$$f(x) = (A + Bx)e^{\alpha x}$$

Where A and B are constants. Setting $f^n(0) = a_n$ gives us:

$$A = a_0$$

Differentiating $f(x)$ gives us:

$$f'(x) = Be^{\alpha x} + \alpha\left(A + Bx\right)e^{\alpha x}$$

So, we can deduce that:

$$a_1 = B + \alpha A \implies B = a_1 - \alpha a_0 \quad (\text{As } A = a_0)$$

This gives us our particular solution of $f(x)$:

$$f(x) = \left(a_0 + \left(a_1 - \alpha a_0\right)x\right)e^{\alpha x}$$

$$\begin{aligned}
\therefore\ f'(x) &= (a_1 - \alpha a_0)e^{\alpha x} + \alpha\left(a_0 + \left(a_1 - \alpha a_0\right)x\right)e^{\alpha x}, \\
f''(x) &= 2\alpha(a_1 - \alpha a_0)e^{\alpha x} + \alpha^2\left(a_0 + \left(a_1 - \alpha a_0\right)x\right)e^{\alpha x}, \\
f'''(x) &= 3\alpha^2(a_1 - \alpha a_0)e^{\alpha x} + \alpha^3\left(a_0 + \left(a_1 - \alpha a_0\right)x\right)e^{\alpha x}.
\end{aligned}$$

This pattern continues allowing us to determine the n-th derivative of $f(x)$:

$$f^n(x) = \alpha^{n-1}e^{\alpha x}\left(n(a_1 - \alpha a_0) + \alpha(a_0 + (a_1 - \alpha a_0)x)\right)$$

As $f^n(0) = a_n$, this allows us to derive our closed form equation for case 2 recurrence relations:

$$a_n = \alpha^{n-1}\left(n(a_1 - \alpha a_0) + \alpha a_0\right) \tag{2}$$

Case 3
I will define case 3 to be the case where two constraints are satisfied:

- $c_1^2 - 4c_2 < 0$ so that the auxiliary equation has complex roots,
- If we let γ and γ^* denote the complex roots of the auxiliary equation, then $\Re(\gamma) \neq 0$.

Therefore, as a solution in the form $f(x) = Ce^{mx}$ would have been assumed, we can determine that the general solution of $f(x)$ is:

$$f(x) = Ae^{\gamma x} + Be^{\gamma^* x}$$

Where A and B are constants. Let $\alpha = \Re(\gamma)$ and $\beta = \Im(\gamma)$, so that $\gamma = \alpha + \beta i$ and $\gamma^* = \alpha - \beta i$. This lets us rearrange the general solution:

$$f(x) = e^{\alpha x}\left(Ae^{\beta i x} + Be^{-\beta i x}\right)$$

Using Euler's identity, this gives us:

$$f(x) = e^{\alpha x}\left((A+B)\cos(\beta x) + i(A-B)\sin(\beta x)\right)$$

As A, B, and i are simply constants, we can just deduce that:

$$f(x) = e^{\alpha x}\left(P\cos(\beta x) + Q\sin(\beta x)\right)$$

Where P and Q are constants. Setting $f^n(0) = a_n$ lets us deduce that:

$$P = a_0$$

Differentiating $f(x)$ using the product rule gives us:

$$f'(x) = e^{\alpha x}((\alpha P + \beta Q)\cos(\beta x) + (\alpha Q - \beta P)\sin(\beta x))$$

Therefore, letting us deduce that:

$$a_1 = \alpha P + \beta Q \implies Q = \frac{a_1 - \alpha a_0}{\beta} \quad (\text{As } P = a_0)$$

Giving us our particular solution to $f(x)$:

$$f(x) = e^{\alpha x}\left(a_0\cos(\beta x) + \frac{a_1 - \alpha a_0}{\beta}\sin(\beta x)\right)$$

To find the n-th derivative of $f(x)$, I will again use Leibniz's theorem. To use this, I will define u and v so that $f(x) = uv$ and find the n-th derivatives of these:

$$u := e^{\alpha x} \implies \frac{d^n u}{dx^n} = \alpha^n e^{\alpha x}$$

$$v := a_0\cos(\beta x) + \frac{a_1 - \alpha a_0}{\beta}\sin(\beta x)$$

As $\frac{d^n}{dx^n}(\sin kx) = k^n \sin\left(kx + \frac{n\pi}{2}\right)$ and $\frac{d^n}{dx^n}(\cos kx) = k^n\cos\left(kx + \frac{n\pi}{2}\right)$, this allows us to determine that:

$$\frac{d^n v}{dx^n} = \beta^n a_0 \cos\left(\beta x + \frac{n\pi}{2}\right) + \beta^n\frac{a_1 - \alpha a_0}{\beta}\sin\left(\beta x + \frac{n\pi}{2}\right)$$

Therefore, giving us the n-th derivative of $f(x)$:

$$f^n(x) = \sum_{r=0}^{n}\binom{n}{r}\cdot\alpha^r e^{\alpha x}\cdot\left(\beta^{n-r}a_0\cos\left(\beta x + \frac{(n-r)\pi}{2}\right) + \beta^{n-r}\frac{a_1 - \alpha a_0}{\beta}\sin\left(\beta x + \frac{(n-r)\pi}{2}\right)\right)$$

As $f^n(0) = a_n$, this allows us to derive our closed form equation for case 3 recurrence relations:

$$a_n = \sum_{r=0}^{n}\frac{n!}{r!(n-r)!}\cdot\alpha^r\cdot\left(\beta^{n-r}\left(a_0\cos\left(\frac{(n-r)\pi}{2}\right) + \frac{a_1 - \alpha a_0}{\beta}\sin\left(\frac{(n-r)\pi}{2}\right)\right)\right) \tag{3}$$

Case 4

I will define case 4 to be the case where two constraints are satisfied:

- $c_1^2 - 4c_2 < 0$ so that the auxiliary equation has complex roots,
- If we let γ and γ^* denote the complex roots of the auxiliary equation, then $\Re(\gamma) = 0$.

Therefore, as a solution in the form $f(x) = Ce^{mx}$ would have been assumed, we can determine that the general solution of $f(x)$ is:

$$f(x) = Ae^{\gamma x} + Be^{\gamma^* x}$$

Where A and B are constants. Let $\beta = \Im(\gamma)$ so that $\gamma = \beta i$ and $\gamma^* = -\beta i$. This lets us rearrange the general solution:

$$f(x) = Ae^{\beta i x} + Be^{-\beta i x}$$

Using Euler's identity, this gives us:

$$f(x) = (A + B)\cos(\beta x) + i(A - B)\sin(\beta x)$$

As A, B, and i are simply constants, we can just deduce that:

$$f(x) = P\cos(\beta x) + Q\sin(\beta x)$$

Where P and Q are constants. Setting $f^n(0) = a_n$ lets us deduce that:

$$P = a_0$$

Differentiating $f(x)$ gives us:

$$f'(x) = \beta Q\cos(\beta x) - \beta P\sin(\beta x)$$

Therefore, implying that:

$$a_1 = \beta Q \implies Q = \frac{a_1}{\beta}$$

Giving us our particular solution:

$$f(x) = a_0\cos(\beta x) + \frac{a_1}{\beta}\sin(\beta x)$$

As $\frac{d^n}{dx^n}(\sin kx) = k^n \sin\left(kx + \frac{n\pi}{2}\right)$ and $\frac{d^n}{dx^n}(\cos kx) = k^n \cos\left(kx + \frac{n\pi}{2}\right)$, this allows us to determine that:

$$f^n(x) = \beta^n a_0 \cos\left(\beta x + \frac{n\pi}{2}\right) + \beta^n \frac{a_1}{\beta}\sin\left(\beta x + \frac{n\pi}{2}\right)$$

As $f^n(0) = a_n$, this allows us to derive our closed form equation for case 4 recurrence relations:

$$a_n = \beta^n \left(a_0 \cos\left(\frac{n\pi}{2}\right) + \frac{a_1}{\beta}\sin\left(\frac{n\pi}{2}\right)\right) \tag{4}$$

2.2.2 Final formulas

Following the derivation of these specific case formulas, I will rewrite them, so that they can be defined concisely. I will list the constraints that are required to be satisfied to use a particular formula, indicating which case-specific formula should be used. I will also detail the values of any parameters within these formulas that have not been previously defined. Again, these formulas are used to find closed forms of homogeneous second-order linear recurrence relations with constant coefficients, so the recurrence relations will have the form:

$$a_n = -c_1 a_{n-1} - c_2 a_{n-2}$$

Where c_1 and c_2 are constants, $c_2 \neq 0$, and the initial values of a_1 and a_2 have been specified. However, I will use a_0 in the formulas, which I will define:

$$a_0 = -\frac{a_2 + c_1 a_1}{c_2}$$

The corresponding auxiliary equation will have the form:

$$m^2 + c_1 m + c_2 = 0$$

Case 1
Constraint(s):

- $c_1^2 - 4c_2 > 0$

Formula:

$$a_n = \frac{1}{\alpha - \beta}\left(\left(a_1 - \beta a_0\right)\alpha^n + \left(\left(\alpha a_0 - a_1\right)\beta^n\right)\right. \tag{1}$$

Where:

- $\alpha = \frac{-c_1 + \sqrt{c_1^2 - 4c_2}}{2}$
- $\beta = \frac{-c_1 - \sqrt{c_1^2 - 4c_2}}{2}$

Case 2
Constraint(s):

- $c_1^2 - 4c_2 = 0$

Formula:

$$a_n = \alpha^{n-1}\left(n(a_1 - \alpha a_0) + \alpha a_0\right) \tag{2}$$

Where:

- $\alpha = -\frac{c_1}{2}$

Case 3
Constraint(s):

- $c_1^2 - 4c_2 < 0$,
- If γ and γ^* are the complex roots of the auxiliary equation, then $\Re(\gamma) \neq 0$.

Formula:

$$a_n = \sum_{r=0}^{n} \frac{n!}{r!(n-r)!} \cdot \alpha^r \cdot \left(\beta^{n-r}\left(a_0 \cos\left(\frac{(n-r)\pi}{2}\right) + \frac{a_1 - \alpha a_0}{\beta} \sin\left(\frac{(n-r)\pi}{2}\right)\right)\right) \tag{3}$$

Where:

- $\alpha = -\frac{c_1}{2}$
- $\beta = \frac{\sqrt{\left|c_1^2 - 4c_2\right|}}{2}$

Case 4
Constraint(s):

- $c_1^2 - 4c_2 < 0$,
- If γ and γ^* are the complex roots of the auxiliary equation, then $\Re(\gamma) = 0$.

Formula:

$$a_n = \beta^n \left(a_0 \cos\left(\frac{n\pi}{2}\right) + \frac{a_1}{\beta} \sin\left(\frac{n\pi}{2}\right) \right) \tag{4}$$

Where:

- $\beta = \frac{\sqrt{|c_1^2 - 4c_2|}}{2}$